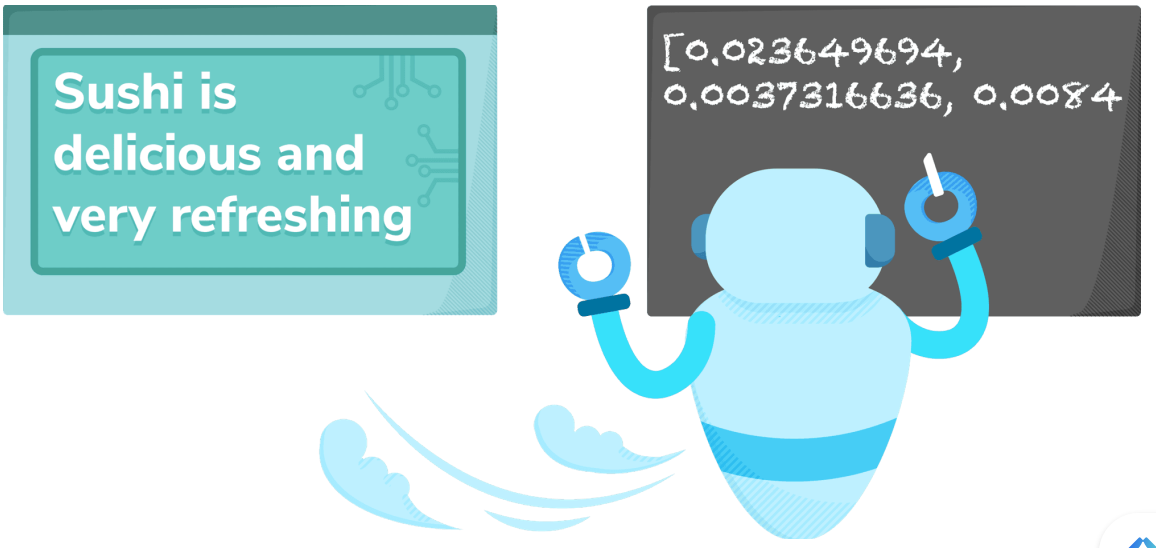
Embeddings

Explore how to use the OpenAI embeddings endpoint to transform text into numerical vectors for machine learning. Learn to calculate semantic similarity using cosine similarity and apply embeddings in practical NLP applications like anomaly detection and text clustering.

The embeddings endpoint

Embedding is a method to represent the data in a vector of continuous numbers. We can provide these vectors to machine learning algorithms and models. Similar texts will have the same embedding vectors, and two different texts will have very different embeddings. OpenAI API takes text as input and returns the embedding vector.



Embeddings

Ask

The Embeddings API call

To get an embeddings vector for a chunk of text, we can call the following function:

```
response = client.embeddings.create(
    input="The text whose embeddings are required",
    model="<model_name>"
)
```

Understanding the embeddings endpoint

Let's look at the embeddings endpoint in more detail, reviewing the request parameters and the response parameters.

Request parameters

Let's look at the parameters that are required to make a request at the embeddings endpoint.

Fields	Format	Type	Descriptior
`model`	String	Required	The ID of the engine to use for this
`input`	String/Array	Required	Input text for embedding, encoded array of tokens. For embedding mu single request, provide an array co arrays of tokens. The input size sh maximum input token limit for the r `text-embedding-ada-002`). Additic an empty string, and any dimensions.

Ask

`encoding_format`	String	Optional	The format in which embeddings is format can either be `float` or `bas`
`dimensions`	Integer	Optional	The specified number of dimensior output embeddings. This feature is from `text-embedding-3` models a
`users`	String	Optional	A unique identifier that represents OpenAI in monitoring and identifyir abuse.

> **Note**: You can learn more about the `model` argument in this lesson.

Response fields

Let's see the important response fields below:

Fields	Format	Description
`index`	Integer	The index of the embedding in the list of en
`embedding`	Array	The embedding vector, which is a list of flo
`object`	String	The object type, which is always `embeddin`

The input to this endpoint is an input text. The output is an array that contains the vector for the text. The embedding vector can be extracted as `response.data[0].embedding` from the response object.

Ask

Embeddings: text-in and embeddings-out model

Converting text to vector

Now that we understand that the embeddings endpoint can convert text to vectors, let's use it to convert text to vectors.

Example

Let's see an example of the embeddings endpoint.

Please provide values for the following:

SECRET_KEY Not Specified...

Edit Import Values from JSON

main.py Python

```python
from openai import OpenAI
client = OpenAI(api_key="{{SECRET_KEY}}")

response = client.embeddings.create(
    input="The burger was fantastic.",
    model="text-embedding-ada-002"
)
```

Ask

```
print(response.data[0].embedding)
```

Run

Finding text embeddings

Let's see some code details:

- **Line 1:** We import the `OpenAI` library.
- **Line 2:** We create an OpenAI client instance using the provided API key.
- **Line 4:** We initiate an embeddings request.
- **Line 6:** We specify the model to be used for generating responses.
- **Line 12:** We print the content of the first choice of response from the generated embeddings.

Cosine similarity of embeddings

To find quantifiable relevance between two pieces of text, we can use the cosine similarity. To do that, we need to convert the text into numerical form. The embeddings vectors retrieved from the embeddings endpoint can be directly used to calculate the cosine similarity. In the code widget below, we generate embeddings of two sentences and use those embeddings to find the similarity score.

Cosine similarity measures the cosine of the angle between two vectors in a high-dimensional space, indicating their similarity. A high cosine similarity value close to 1 signifies that the vectors are pointing in similar directions, implying semantic similarity in NLP tasks. Conversely, a low cosine similarity value near zero suggests that the vectors are orthogonal or divergent, indicating dissimilarity in meaning between the words or phrases they represent.

Ask

Example

Let's look at an example of the embeddings endpoint being used to measure the similarity in two statements.

Please provide values for the following:

SECRET_KEY Not Specified...

Edit Import Values from JSON

main.py Python

```
from openai import OpenAI
from scipy.spatial.distance import cosine

client = OpenAI(api_key="{{SECRET_KEY}}")

# Get embeddings for two statements
response = client.embeddings.create(
    input=["The burger was fantastic.", "I really enjoyed the burger."],
    model="text-embedding-ada-002"
)

# Extract embeddings
embedding1 = response.data[0].embedding
```

Run

Finding similarity between text samples

Practical use case

OpenAI's embeddings endpoint can be employed for a variety of real-world applications across different industries and domains. Some examples incl

Ask

- **Anomaly detection**: In applications where anomalies need to be detected in text data (e.g., identifying fraudulent transactions or abnormal behavior in

user interactions), embeddings can be used to represent normal patterns. Unusual or anomalous text inputs can then be identified by measuring the distance between their embeddings and the embeddings of normal data points.

- **Text similarity and clustering**: By obtaining embeddings for various text

Ask